

▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 12 summarizes the current fairness state for upload 12. Accuracy changed by +0.750 and fairness score changed by +0.973. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.884.

ACCURACY

0.750

▲ +0.750

FAIRNESS SCORE

0.973

▲ +0.973

COMBINED SCORE

0.884
ThresholdOptimizer_mitigated_threshold

DISPARATE IMPACT

0.824

Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	german_log_reg_baseline.pkl	comparison

Fairness Metrics

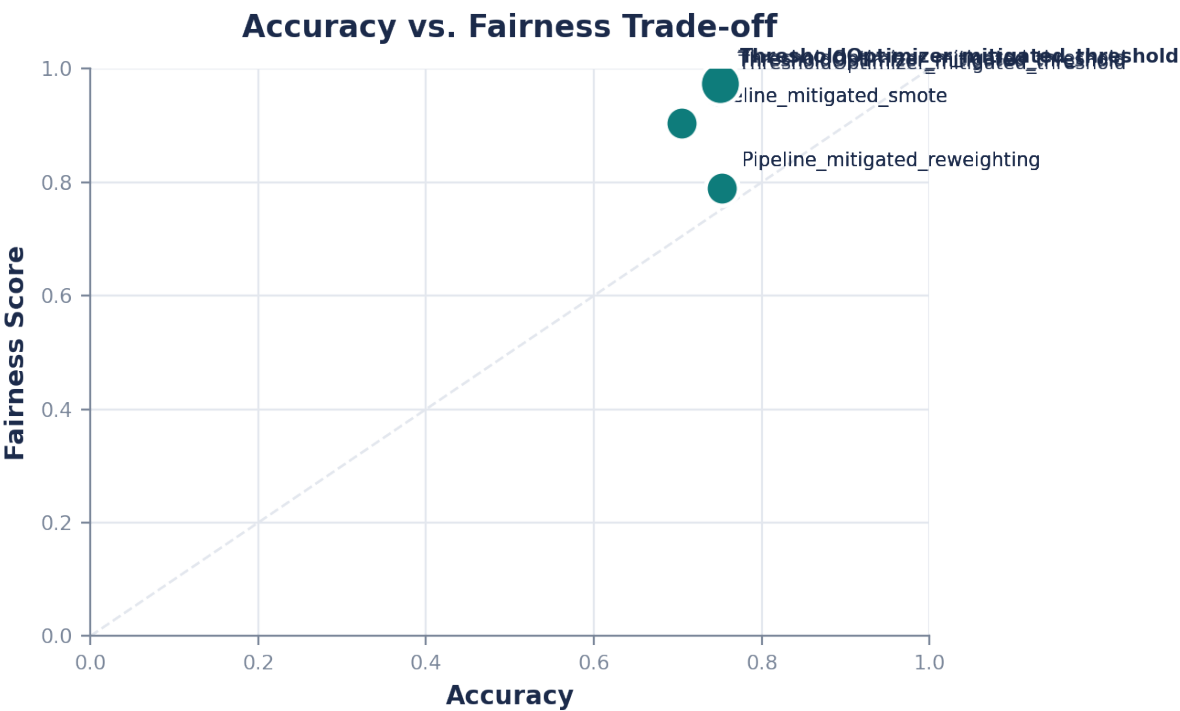
METRIC	VALUE	INTERPRETATION
Demographic Parity Difference (DPD)	0.041	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.012	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.824	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

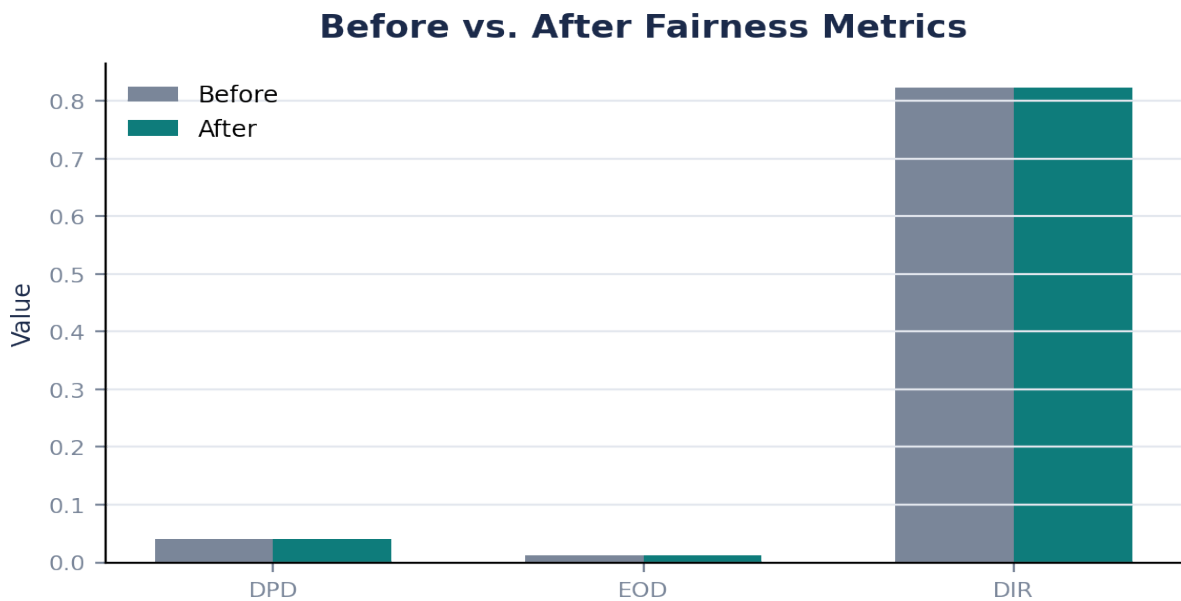
Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.750	0.973	0.041	0.012	0.824	0.884

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.748	0.970	0.022	0.039	0.899	0.881
★ ThresholdOptimizer_mitigated_threshold	0.747	0.967	0.035	0.031	0.849	0.879
★ ThresholdOptimizer_mitigated_threshold	0.748	0.961	0.046	0.032	0.806	0.876
Pipeline_mitigated_smote	0.705	0.905	0.104	0.087	0.796	0.825
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Pipeline_mitigated_smote	0.705	0.905	0.104	0.087	0.796	0.825
Pipeline_mitigated_reweighting	0.752	0.790	0.159	0.260	0.469	0.775
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Accuracy vs. Fairness Trade-off



Before vs. After Fairness Metrics



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.